Swami Keshvanand Institute of Technology, Management & Gramothan, Jaipur

B. Tech. I Year I Semester 2024-25

Assignment Sheet-II
Session 2024-25
(Odd Semester)

Subject: Computational Thinking and Programming **Max Marks- 60**

Due Date: 03-12-24

Roll No.	1-6	7-12	13-18	19-24	25-30	31-36	37-42	43-48	49-54	55-60	61-66
Set-Number	I	II	III	IV	V	VI	VII	VIII	IX	X	XI

(Set-I)

Q. No	Question	BL	CO	MM
1	What do understand by String and how it is different from character array.	1	3	10
2	Write a program to find out sum of each row and sum of each column of a 2D-array.	2	3	10
3	What is function? why we use function in C? Explain with example.	1	4	10
4	Write a program to find out factorial of a number using recursion	2	4	10
5	Compare structure and union	2	5	10
6	Develop a structure namely Book and create an array of Book structure with size 10.	2	5	10

Swami Keshvanand Institute of Technology, Management & Gramothan, Jaipur

B. Tech. I Year I Semester 2024-25

Assignment Sheet-II

Session 2024-25

(Odd Semester)

Subject: Computational thinking and Programming. **Max Marks- 60**

(Set-II)

Q. No	Question	BL	CO	MM
1	What is array ? Explain the declaration and initialization of one dimensional and two dimensional array with an example	1	3	10
2	Write a C program to read N integers into an array A and to find the (i)sum of odd numbers,(ii) sum of even numbers,(iii) average of all numbers. Output the results computed with appropriate headings	2	3	10
3	Write a C function isprime(num) that accepts an integer argument and returns 1 if the argument is a prime or 0 otherwise. Write a program that invokes this function to generate prime numbers between the given ranges	2	4	10
4	What is pointer? How is a pointer initialized?	1	4	10
5	Explain in brief with example: a) declaring structure variables b) accessing structure variables c) size of structure	1	5	10
6	Write a program that reads a file containing integers and appends at its end the sum of all the integers.	2	5	10

Swami Keshvanand Institute of Technology, Management & Gramothan, Jaipur

B. Tech. I Year I Semester 2024-25

Assignment Sheet-II

Session 2024-25

(Odd Semester)

Subject: Computational thinking and Programming. **Max Marks- 60**

(Set-III)

Q. No	Question	BL	CO	MM
1	Write a C program that accepts (3×3 ordered matrices A and B),and compute the following (i)summation of Two numbers (ii) Subtraction of Two numbers	1	3	10
2	Write a C program to check whether the given string is palindrome or not without using in built function	2	3	10
3	What is the principal difference between the functions malloc and calloc?	1	4	10
4	Write a program in „C“ using functions to swap two numbers	2	4	10
5	What is structure? How it is differed from an array.	1	5	10
6	Write a C program to count no.of characters, spaces, lines, words of a file	2	5	10

Swami Keshvanand Institute of Technology, Management & Gramothan, Jaipur

B. Tech. I Year I Semester 2024-25

Assignment Sheet-II

Session 2024-25

(Odd Semester)

Subject: Computational thinking and Programming. **Max Marks- 60**

(Set-IV)

Q. No	Question	BL	CO	MM
1	Define string. How string is declared and initialized? Explain string input/output functions with an example	1	3	10
2	Write a C Program to implement string copy operation STRCOPY(str1,str2)	2	3	10

3	Given the following declarations:
	int x=10, y=10;
	int *p1 = &x, *p2 = &y;
	What is the value of each of the following expressions?
	a) (*p1)++
	b) –(*p2)
	c) *p1 + (*p2)--
	d) ++(*p2)- *p1
		2	4	10

4	Give the scope and life-time of the following (i)External variable (ii) Static variable (iii) Automatic variable (iv) Register variable	1	4	10
5	How does an append mode differ from a write mode?	1	5	10
6	Write a C program to count no.of characters, spaces, lines, words of a file	2	5	10

Swami Keshvanand Institute of Technology, Management & Gramothan, Jaipur

B. Tech. I Year I Semester 2024-25

Assignment Sheet-II

Session 2024-25

(Odd Semester)

Subject: Computational thinking and Programming. **Max Marks- 60**

(Set-V)

Q. No	Question	BL	CO	MM
1	What do understand by String and how it is different from character array.	1	3	10
2	Write a program to perform linear search on an array of size 10.	2	3	10
3	What is recursion function? explain with example	1	4	10
4	What is dangling pointer? Explain with example	1	4	10
5	Explain in brief with example: a) declaring structure variables b) accessing structure variables c) size of structure	1	5	10
6	Write a program that reads a file containing integers and appends at its end the sum of all the integers	2	5	10

Swami Keshvanand Institute of Technology, Management & Gramothan, Jaipur

B. Tech. I Year I Semester 2024-25

Assignment Sheet-II

Session 2024-25

(Odd Semester)

Subject: Computational thinking and Programming. **Max Marks- 60**

(Set-VI)

Q. No	Question	BL	CO	MM
1	What is array ? Explain the declaration and initialization of one dimensional and two dimensional array with an example	1	3	10
2	Write a C program to find the largest element in an array	2	3	10
3	Give the scope and life-time of the following (i)External variable (ii) Static variable (iii) Automatic variable (iv) Register variable	1	4	10
4	Write a C function isprime(num) that accepts an integer argument and returns 1 if the argument is a prime or 0 otherwise. Write a program that invokes this function to generate prime numbers between the given ranges	2	4	10
5	Write a C program to count no.of characters, spaces, lines, words of a file	2	5	10
6	Create a structure namely Book and an array of Book structure with size 10.	2	5	10

Swami Keshvanand Institute of Technology, Management & Gramothan, Jaipur

B. Tech. I Year II Semester 2023-24

Assignment Sheet-II

Session 2024-25

(Odd Semester)

Subject: Computational thinking and Programming. **Max Marks- 60**

(Set-VII)

Q. No.	Question	BL	CO	MM
1	What is a void pointer, and how is it different from other types of pointers? Provide examples of its usage in C.	2	4	10
2	Explain the difference between predefined and user-defined functions in C with suitable examples.	2	4	10
3	Create a program that concatenates two strings without using predefined library functions. Justify the use of pointers in your implementation.	6	3	10
4	Design a program to concatenate two strings without using predefined library functions. Use pointers for accessing string elements.	6	3	10
5	Evaluate the pros and cons of using pointers with structures for dynamic memory management in C.	4	5	10
6	Demonstrate how a structure can be passed to a function in C, and how modifications to its members can be made within the function using pointers.	3	5	10

Swami Keshvanand Institute of Technology, Management & Gramothan, Jaipur

B. Tech. I Year II Semester 2023-24
Assignment Sheet-II
Session 2024-25
(Odd Semester)

Subject: Computational thinking and Programming. **Max Marks- 60**

(Set-VIII)

Q. No.	Question	BL	CO	MM
1	Write a program to demonstrate call by value and call by reference using functions. What are the key differences observed in the outputs?	3	4	10
2	Evaluate the impact of improper pointer arithmetic on a program's memory access and explain how such issues can be avoided.	5	4	10
3	Design a function that accepts a one-dimensional integer array as a parameter, finds the largest and smallest elements, and returns them. Evaluate the advantages of passing the array by reference.	5	3	10
4	Compare the use of pointers with arrays for accessing string elements. What are the advantages and limitations of each approach?	4	3	10
5	Explain the difference between passing an array and passing a string as a parameter to a function in C.	2	3	10
6	Compare the use of structures and unions for storing data. Under what conditions would you prefer a union over a structure?	4	5	10

Swami Keshvanand Institute of Technology, Management & Gramothan, Jaipur

B. Tech. I Year II Semester 2023-24

Assignment Sheet-II

Session 2024-25

(Odd Semester)

Subject: Computational thinking and Programming. **Max Marks- 60**

(Set IX)

Q. No.	Question	BL	CO	MM
1	Explain with examples how pointer arithmetic can be used to navigate through an array in C. How does this differ from using array indexing?	4	4	10
2	Design a C program that dynamically allocates memory for an array using a void pointer, performs arithmetic operations on the elements, and then deallocates the memory.	6	4	10
3	Analyze the memory layout of a two-dimensional array in C when accessed using row-major and column-major order. How does it affect performance?	4	3	10
4	Write a C program to find the sum of elements in a two-dimensional array using nested loops.	3	3	10
5	Analyze the behavior of file handling functions in C when attempting to read from or write to a file that does not exist. How can such errors be handled programmatically?	4	5	10
6	Write a C program to read the contents of a text file and count the number of words in it.	3	5	10

Swami Keshvanand Institute of Technology, Management & Gramothan, Jaipur

B. Tech. I Year II Semester 2023-24

Assignment Sheet-II

Session 2024-25

(Odd Semester)

Subject: Computational thinking and Programming. **Max Marks- 60**

(Set X)

Q. No.	Question	BL	CO	MM
1	Implement a recursive function to calculate the factorial of a number. Compare this approach with an iterative implementation in terms of memory usage and performance.	5	4	10
2	Write a C program that uses a recursive function to calculate the factorial of a number.	3	4	10
3	Write a program to reverse a string using pointer variables. Test the program with different inputs and explain the process.	3	3	10
4	Define a one-dimensional array and explain how it is declared and initialized in C.	1	3	10
5	Write a C program to read the contents of a text file and count the number of words in it.	3	5	10
6	Evaluate the pros and cons of using pointers with structures for dynamic memory management in C.	5	5	10

Swami Keshvanand Institute of Technology, Management & Gramothan, Jaipur

B. Tech. I Year II Semester 2023-24

Assignment Sheet-II

Session 2024-25

(Odd Semester)

Subject: Computational thinking and Programming. **Max Marks- 60**

(Set-XI)

Q. No.	Question	BL	CO	MM
1	Develop a program that uses a pointer to a pointer to dynamically allocate memory for a 2D matrix. Explain why this method is preferred over static allocation in certain scenarios.	6	4	10
2	Compare the use of pointers with arrays for accessing string elements. What are the advantages and limitations of each approach?	4	3	10
3	Explain the difference between one-dimensional and two-dimensional arrays in C. Provide a real-world application where each is used effectively	2	3	10
4	Analyze the advantages and disadvantages of using pointers for parameter passing in functions compared to call by value. Provide scenarios where each method is preferable.	4	4	10
5	Design a program in C to manage student records using structures and file handling. The program should allow adding, searching, and updating records stored in a file.	6	5	10
6	Define a structure and a union in C. How do they differ in terms of memory allocation?	1	5	10